

includes support for CUDA, OpenCL, and the fracture of rigid bones.

The library is a cross-platform library and supports Playstation 3, Xbox 360, Wii, PCs with the Linux and Mac OSX, and the iPhone. It has a very advanced feature set:

- Discrete and continuous collision detection including ray and convex sweep test. Support for all basic primitive shapes and concave/convex meshes.
- Fast and stable rigid body dynamics, vehicle dynamics, character control and slides, hinge physics, plus twist constraint for ragdoll physics.
- Soft body dynamics for cloth, hair/fur, rope objects and deformable volumes, supporting two-way interaction with rigid bodies plus binary constraints.
- Integration with major development tools such as Maya, Blender and COLLADA.
- Completely written in C++ and the source has been open sourced under the Zlib License.

As Bullet is fairly unknown in the general gamer circles (unlike Blender, OGRE and Open AL), you may be wondering which projects have used it. How about: *Grand Theft Auto 4*, Sony's *Free Realms*, and *2012* (last year's Hollywood blockbuster). Bullet Physics is also included within the RAGE Engine of Rockstar Games.

RakNet

RakNet is a cross-platform, open source, object-oriented networking library for game programmers, written in C++ by RakkarSoft. A lower-level framework powers the higher-level features such as game object replication, voice chat, and patching binaries.

It is a DirectPlay alternative touted as being better than the DirectX component due to its simpler integration. The library uses UDP for all network connections and is written primarily for game development. The library is available in three different licensing models: GPL, shareware and commercial licences. RakNet also offers free licences for indie developers.

The major features of the library include:

- Object replication: Automatically creates, destroys, serialises and transmits game objects
- Player lobby: Database-powered lobby with support for friends, rooms, quick matches, and ranking
- Secure connections: Support for SHA1, AES128, SYN cookies and RSA
- A robust communication layer with automatic flow control, message ordering on multiple channels, message coalescence, and splitting and reassembly of packets.
- An autopatcher to automatically patch the software at the gamer's end with the latest updates
- Support for the various C/C++ remote procedure calls.
- Support for voice communication, including bindings for Port Audio, FMOD, and DirectSound, besides NAT Punchthrough for voice support over P2P applications.

As it's also used as a commercial tool, most applications of RakNet are protected under NDAs. So a finite list of games using RakNet is not freely available. However, RakNet is being used in a number of games by *Epic Games China*, *Codemasters*, and the *Unity Engine*. It was also used in the games *Crashday* and *Hyperball Racing*.

In a nutshell

Yes, yes, the tools look totally awesome, we know... and they sound jam packed with features and efficiency. Yes! You will be surprised with how frequently they are used in major titles. We suggest you check out the *Further Reading* section and try out these tools for yourself.

Step up and reach out—all the awesomeness is within reach. Sing the song!

Further Reading

- OGRE 3D Website: <http://www.ogre3d.org/>
- OGRE 3D Features: <http://www.ogre3d.org/about/features>
- The Story of OpenAL: <http://www.linuxjournal.com/article/4400>
- About Open AL - <http://connect.creativelabs.com/openal/default.aspx>
- List of Games using Open AL - <http://connect.creativelabs.com/openal/OpenAL%20Wiki/Games.aspx>
- Applications of Open AL (Wikipedia) - <http://en.wikipedia.org/wiki/OpenAL#Applications>
- Blender Website: <http://www.blender.org/>
- Blender 3D Features - <http://www.blender.org/features-gallery/features/>
- Community of Blender Artists: <http://blenderartists.org/>
- Bullet Physics Features List: http://bulletphysics.com/Bullet/BulletPhysics2.74_2.htm#_Toc223854815
- Bullet Physics on Google Code: <http://code.google.com/p/bullet/>
- The Bullet Physics Issue List - <http://code.google.com/p/bullet/issues/list>
- Working with Bullet Physics: http://bulletphysics.com/Bullet/BulletPhysics2.74_2.htm
- RakNet Features List: <http://www.jenkinsoftware.com/raknet/index.html>
- Visual Guide to RakNet Features: <http://www.rakkarsoft.com/raknet/manual/RakNetMajorFeatures.html>
- RakNet Licensee List: <http://www.jenkinsoftware.com/licensees.html>

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